

Instructions: Answer all the questions from Part I and any two from Part II
(The right side numbers indicate marks.)

Part I

1. (a) What are drift and diffusion currents through a semiconductor? 4
(b) An extrinsic Si sample is made by adding a donor type impurity to the extent of 1 atom per 10^8 Si atoms of an intrinsic semiconductor (Si) at 300K. A steady state current of $2\mu\text{A}$ exists in the bar made up of the extrinsic Si sample. Determine (i) the conductivity and (ii) the voltage across the bar having length of 3mm and a rectangular cross-section of $50\mu\text{m} \times 10\mu\text{m}$. 6
(Given, intrinsic carrier concentration (n_i) = $1.45 \times 10^{10}/\text{cc}$, concentration of Si atoms = 5×10^{22} atoms/cc, mobility of an electron (μ_n) = $1500 \text{ cm}^2/\text{V-s}$, mobility of a hole (μ_p) = $500 \text{ cm}^2/\text{V-s}$ for Silicon at 300K)
2. (a) Sketch the characteristics for an ideal diode and approximate characteristics for a practical Si diode. Briefly explain each of the characteristics. 4
(b) Draw the equivalent circuit of a practical and an ideal Si diode. 3
(c) What is barrier potential of a p-n junction diode? Can you measure the barrier potential of a p-n junction diode by connecting a voltmeter across the diode? Justify your answer. 3
3. (a) Sketch the input and output characteristics for a BJT in common emitter configuration. 4
(b) For the transistor (Si) circuit as shown in Figure 1, the device parameter and circuit components are given as follows $\beta = 100$, $V_{CC} = 15 \text{ V}$, $V_{EE} = -10 \text{ V}$, $R_C = 10 \text{ K}\Omega$, $R_B = 8 \text{ K}\Omega$, $R_E = 12 \text{ K}\Omega$. Determine I_C , I_B , I_E , V_E and V_{CE} . 6

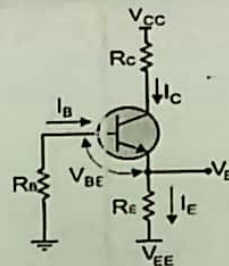


Figure 1

4. (a) Convert $(0.55)_{10}$ and $(10.75)_{10}$ to their corresponding binary equivalent. 5
(b) Convert $(55)_8$ and $(15.35)_8$ to their corresponding binary equivalent. 2
(c) Using 2's complement method perform $(0010) - (1010)_2$
(d) Simplify the Boolean expression: $(AB + C + D)(\bar{C} + D)(\bar{C} + D + E)$
(e) $(4)_3 + (7)_3 = (?)_3$
5. (a) Draw a biased positive clipper circuit using Si diode and a reference voltage of +2 V. Draw the output waveform for a sinusoidal input with a peak to peak value of 10 V. 6
(b) Draw the truth table and logic diagram for a half adder. 4
6. (a) What is CMRR? In a practical op-amp $A_D = 100$ and $A_C = 0.1$ and is operated with input signal voltages of $1050 \mu\text{V}$ and $950 \mu\text{V}$. Find its output voltage and CMRR. 5
(b) Let the magnitude of gain in the inverting op-amp circuit as shown in Figure 2, is X, when switch S is open. Find the magnitude of gain when the switch S is closed in terms of X. 5

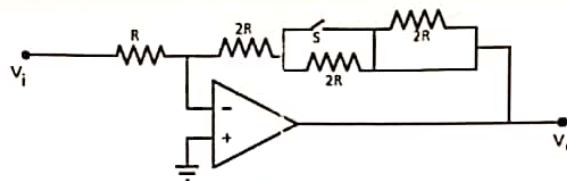


Figure 2

Part II

- 7 (a) What is threshold voltages V_T ? Draw an N channel enhancement type MOSFET showing the source, drain and gate. Explain briefly how does it work and draw the I_D vs V_{DS} characteristic curve. 12
- (b) A full-wave rectifier using double diodes has load resistance $R_L = 1\text{K}\Omega$. Each diode has a forward resistance of 10Ω . The voltage across the secondary winding is $440 \sin(314t)$ Volt. Find 8
- DC value of the load voltage and current.
 - RMS value of the load voltage and current.
 - Ripple factor
 - Rectification efficiency
 - PIV rating of each diode
 - Frequency of the rectified output waveform at the load.
- 8 (a) Draw emitter feedback bias circuit in common emitter configuration with input and output coupling capacitors. Explain the circuit and derive the expression for stability factor in terms of the device parameters and circuit components. 9
- (b) Explain how the stability of emitter feedback circuit as compared to the fixed bias circuit is improved. 3
- (c) Consider a collector emitter feedback bias circuit (Figure 3) with the following specifications: $R_C = 1\text{K}\Omega$, $R_B = 400\text{K}\Omega$, $R_E = 2.2\text{K}\Omega$, $V_{CC} = 15\text{V}$, $\beta = 100$. Determine I_C , I_B , I_E , V_E , V_B , V_{CE} and stability factor $S(I_{CO})$ using the approximate analysis. 8

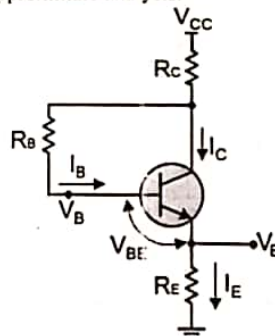


Figure 3

- 9 Simplify the given Boolean expressions using Boolean algebra and also by 4-variable K-map method. 20
- $f = \sum m(1, 3, 4, 6, 9, 11, 12, 15)$; (into SOP form)
 - $f = \prod M(0, 4, 6, 7, 8, 12, 13, 14, 15)$; (into POS form)